Invertebrate Abundance Lab

Data Analysis

Download the class data as a CSV and open it in JASP.

Descriptives → Split by microhabitat

Report the mean and standard deviation of total invertebrate count for each microhabitat (1 point).

Display a boxplot of the data. Which microhabitat appears to have the highest median abundance? Which appears to have the most variability (or the widest spread)? (2 points)

ANOVA → Request homogeneity test and q-q plot

Is the assumption of homogeneity of variance met? Explain. (2 points)

Based on the Q-Q plot (and given that we have count data and will see "steps" on the plot), do the residuals look approximately normally distributed? If not, what shape of departure do you see (e.g., upward curve on right side/right-skew, etc.)? (1 point)

Count data like this is a known case where ANOVA assumptions are often violated. If you feel it necessary, you can perform a square root transformation on the invertebrate abundance data by computing a column with this R code: sqrt(Invertebrate Abundance+1)

Report the F-statistic and the p-value of your ANOVA. Do you reject the null hypothesis? State the null as a part of your answer. (2 points)

If your ANOVA was significant, display a Tukey HSD post-hoc test. Which specific pairs of microhabitats differ significantly from each other ($p < 0.05$), and which do not? (If your ANOVA was not significant, please restate that here.) (1 point)

What is the experimental unit in this study? Briefly explain. (1 point)

Propose at least one ecological explanation for the pattern you found (or didn't find) across microhabitats. (2-3 sentences) (1 point)

For most of our microhabitats, we used a fixed time (2.5 minute search) + fixed area (quadrat) approach to sampling for invertebrates. What abstract concept do these kinds of things help control for, and why might ecologists want to specify these when conducting a repeatable study? (2-3 sentences) (1 point)

What is at least one source of bias/confounding in this design that could produce a false microhabitat effect (even if our ANOVA was not significant/invertebrates don't actually prefer one habitat over another)? (Think about space, time, etc.) (2-3 sentences) (1 point)

Deliverables

A PDF or otherwise of your written answers to the questions, and your JASP results (PDF).

This will be due on Friday, October 16th, at 11:59 PM.